TOPIC 2 · CREATE SKILLS FOR MICROSOFT 365 COPILOT
LAB 4 THEME · REPEATABLE REPORTING

Lab 4 — Use and Improve the Skill in Excel and PowerPoint

Maps to	LO2: create and improve a reusable maintenance skill in Excel and PowerPoint
Duration	40 minutes
Tools	Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive
Mock data	data/Northstar_Work_Orders.xlsx
Lab files	data/templates/maintenance-reporting/SKILL.md (Complete custom maintenance skill template)
Course	Microsoft 365 Copilot Masterclass (C27) · v2.1
Prompts	https://tertiarycourses.github.io/C27-Microsoft-365-Copilot-Masterclass/prompts/#lab-4

Goal

Northstar_Skill_Report.xlsx with Maintenance_Summary and Assignment_Proposals, Northstar_Skill_Update.pptx, SKILL.md v2 and a before/after improvement log.

What You Will Do

Use the Lab 3 skill to create a verified summary and improve the initial PowerPoint update. Then deliberately test missing inputs and a rerun, tighten the skill to v2, and compare before/after evidence. The target is consistent reporting, not a longer prompt or an unsupported promise of autonomous approval.

What You Will Build

Northstar_Skill_Report.xlsx with Maintenance_Summary and Assignment_Proposals, Northstar_Skill_Update.pptx, SKILL.md v2 and a before/after improvement log.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-c027 is your own learner output folder.
- Lab 3 skill v1 and test record; Lab 2 update, or named supplied checkpoints.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Copy the Lab 3 reviewed SKILL.md to your output folder and retain v1. Open this lab's clean workbook or your Lab 1 cleaned workbook. Save as Northstar_Skill_Report.xlsx. Read Technicians: primary category and available hours are synthetic planning inputs, not approvals. Check that your skill is enabled in Excel.

Step 2. Invoke the skill in Plan or Chat only, review its proposed output boundaries, then allow editing. Inspect the resulting formulas and charts. Summary counts must reconcile to Control, and the source table must remain 120 unique records.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@maintenance-reporting Build Maintenance_Summary for tblWorkOrders at 2026-09-30 with all
filters All. Include total, closed, cancelled, backlog, overdue, on-time closed rate and
recorded cost; show formulas and source scope. Plan first. Preserve source records.

EXCEL - ALLOW EDITING:
Apply that reviewed summary plan. Add Assignment_Proposals for unassigned backlog only. Use
tblTechnicians primary category and available hours as planning inputs; show category/capacity
checks, current owner and proposed owner separately. Leave manager decision UNKNOWN. Do not
change Assigned_To.
```

Step 3. Check the unassigned backlog count against Control. A technician with the same category may still lack usable capacity: use available hours minus known proposed hours, flag conflicts, and do not convert a suggestion into confirmed assignment. Export your verified summary into the Word snapshot template, preserving measures and date.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
@maintenance-reporting Review Assignment_Proposals against tblWorkOrders and tblTechnicians.
Check each suggested owner has the matching water-service job category, capacity limitations
are visible and original Assigned_To values are unchanged. Leave manager approval UNKNOWN; do
not assign or schedule a technician.
```

Step 4. Open the four-slide initial update from Lab 2, or use data/checkpoints/Northstar_Initial_Update_Checkpoint.pptx, a four-slide rejoin example explicitly labelled as a checkpoint. Save as Northstar_Skill_Update.pptx. Select the reviewed snapshot in PowerPoint and invoke your installed skill.

PROMPT / CHECK — copy and adapt

```
POWERPOINT COPILOT:
@maintenance-reporting Improve this existing initial maintenance update from the selected
verified snapshot. Retain four slides and accurate values; strengthen claim headlines,
simplify text, improve source notes and use the current design. Do not duplicate slides or
imply live links to Excel. Label suggested reassignment as proposed.
```

Step 5. Create a new EMPTY workbook with no table, open a fresh Copilot chat and run a missing-input test for a different reporting period with no source selected. A proper skill asks for source/date/scope rather than inventing metrics. If it guesses, record the failure and revise the skill's input guard. Return to Northstar_Skill_Report.xlsx and Save As Northstar_Skill_Rerun_Test.xlsx. Run the second summary request in this copy of the report workbook, not in the empty workbook; only named outputs may be replaced after review, with no duplicated source or output sheets. Compare a before/after copy of Work_Orders sorted by Work_Order_ID for status, owner, dates and cost; every original source value must match.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
@maintenance-reporting Prepare my next maintenance report. I need a DIFFERENT reporting period
from the course example and have not supplied any source, date or scope. This empty workbook
contains no work orders. Tell me what you need; do not invent data.

EXCEL - PLAN:
@maintenance-reporting Rebuild the existing Maintenance_Summary for the same table, date and
scope. Describe what will be replaced, ask before replacement and keep source/unrelated
sheets. Do not duplicate outputs.
```

Step 6. In a plain text editor improve SKILL.md: require an input guard, category/capacity evidence, source provenance per claim and a clear rerun rule. Retain YAML name maintenance-reporting and add a body version note v2. Re-upload/Refresh in BOTH Excel and PowerPoint. Record actual before/after changes, or already present - retained for safeguards that your v1 already contains. Do not fabricate differences. Rerun the smoke tests and compare values. If using the manual route, replace the pasted instructions in both apps.

PROMPT / CHECK — copy and adapt

```
MICROSOFT 365 COPILOT WEB:
Improve this supplied maintenance-reporting skill to v2 using only my actual smoke-test and
rerun findings pasted here. Add explicit missing-input and no-duplicate-output guards. Keep
Excel source and current owners protected. Return the complete revised SKILL.md and a short
before/after change log; do not fabricate test evidence.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/C27-Microsoft-365-Copilot-Masterclass/prompts/#lab-4

#	Purpose	Prompt
1	Excel	@maintenance-reporting Build the verified summary and separate assignment proposals.
2	PowerPoint	@maintenance-reporting Improve the existing four-slide update; preserve facts and count.

Test It

Check	Expected result
Source preserved	120 unique orders; current owner/status unchanged.
Skill outputs	Backlog 60; overdue 50; on-time closed 30/48; cost S$35,827.50.
Improvement evidence	Input guard and rerun tested; v1/v2 retained; PowerPoint four-slide output accurate.

Checkpoint

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?